

Prague University of Economics and Business
Faculty of Informatics and Statistics



Textual-Financial Divergence and Stock Return Volatility: Evidence from U.S. Corporate Filings

Bachelor Thesis

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Author: Šimon Slanský

Supervisor: prof. RNDr. Ing. Michal Černý, Ph.D.

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Abstract

Corporate 10-K filings consist of both audited financial data and extensive narrative sections written by management, yet most empirical models work with these two information channels separately. Whether the *interaction* between narrative language and reported financial metrics has any additional information about firm-level risk is still largely unexplored. This thesis investigates whether a mismatch between the direction of change in managerial language and the direction of financial change—in this thesis called *textual-financial divergence*—helps determine future stock return volatility. I build an unbalanced panel of 537 large U.S. companies over fiscal years 2010–2024 (6,299 firm-year observations in the regression sample) and extract the Risk Factors (Item 1A) and Management’s Discussion (Item 7) sections from each 10-K filing using a custom SEC EDGAR parser. Year-over-year changes in sentiment, risk-word density, and textual similarity are computed and combined with standard financial controls in absorbing OLS regressions with industry and year fixed effects. Results from the financial baseline model show that lagged volatility, firm size, and profitability together explain 65.4% of the cross-sectional variation in post-filing annualised volatility. Lagged volatility is the strongest predictor (coefficient 0.55, $t = 20.7$), while ROA has a strong negative effect (-0.22 , $t = -7.9$) and asset growth enters positively ($+0.03$, $t = 3.1$).

Keywords

textual analysis, 10-K filings, SEC EDGAR, volatility, information uncertainty, narrative divergence, sentiment, panel data, corporate disclosures

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List of Abbreviations

CIK Central Index Key

DV Dependent Variable

EDGAR Electronic Data Gathering,
Analysis, and Retrieval

EPS Earnings Per Share

FE Fixed Effects

FY Fiscal Year

FYE Fiscal Year End

GLM Generalized Linear Model

L&M Loughran and McDonald

NLP Natural Language Processing

OCF Operating Cash Flow

OLS Ordinary Least Squares

ROA Return on Assets

SEC Securities and Exchange Commission

SIC Standard Industrial Classification

XBRL eXtensible Business Reporting
Language

Introduction

1. Literature Review

2. Data

2.1 Selecting the Sample

We start with the sample of 1,000 largest U.S. public companies by market capitalisation, taken from the SEC EDGAR company-tickers file, and apply three filters. First, we keep only entities the SEC classifies as “operating,” which removes exchange-traded funds, closed-end funds, blank-cheque companies, and shell entities. Second, we exclude firms in SIC codes 6000–6999 (finance, insurance, and real estate), following common sample-selection practice in prior accounting research (Francis et al., 2005; Sloan, 1996). Third, we require at least five years of readable XBRL data from 10-K filings and 90% valid stock-return coverage, so that there is enough within-firm variation for fixed-effects estimation.

Sample size. We deliberately focus on large, liquid firms rather than the full EDGAR universe for several reasons. First, our regressions use firm fixed effects with standard errors clustered by firm. As Petersen (2009) shows, the reliability of cluster-robust standard errors depends on the number of clusters—here 537—not on total observations, and 537 clusters are well above the threshold for reliable inference. Also, the textual-analysis part of the thesis requires downloading, parsing, and scoring every 10-K in the sample. Expanding to several thousand firms would greatly increase computation time. Lastly, smaller firms tend to have shorter XBRL histories (mandatory electronic filing was phased in by firm size), more tagging errors, and thinner trading, all of which add noise to financial ratios and return-based variables.

The sample size is in line with prior work combining textual and financial analysis of annual filings. Campbell et al. (2014) study about 9,000 firm-years of risk-factor disclosures from large U.S. firms; Loughran and McDonald (2011) use a broader set of roughly 50,000 firm-years, but their goal—validating a sentiment dictionary—calls for maximal cross-sectional coverage, whereas our divergence measure is estimated within firms over time.

2.2 Choosing the Filing Frequency — why annual 10-Ks?

Before moving to variable construction, it is worth explaining why this study uses annual 10-K filings instead of quarterly 10-Q reports. Quarterly data would roughly triple the sample, but two practical problems—shown in Tables 2.1–2.2—ruled out that approach.

Financial data gaps in Q4. 10-Q filings cover only Q1–Q3, so any quarterly panel must reconstruct Q4 as the difference between the annual figure and the sum of the three preceding quarters. Table 2.1 shows that this residual approach is not usable for operating cash flow: the

three quarterly facts coexist in only 7% of firm-years, because most firms report cumulative year-to-date cash flows rather than standalone quarterly amounts. Operating cash flow is the primary input into our OCF-to-assets ratio, so a quarterly panel would lose the cash-flow-based robustness check at the source. Even for net income and operating income, where coverage exceeds 80%, the implied Q4 disagrees with the explicit Q4 by more than one percent of the annual total in about 10–14% of comparable firm-years.

Table 2.1: Q4 Reliability: Availability and Consistency of Fourth-Quarter XBRL Facts

Metric	Firm-Years	Q1 – Q3 (%)	Explicit Q4 (%)	Comparable	Match (%)
Net Income	451	80.9	43.0	172	89.5
Operating Income	391	79.0	22.0	72	86.1
Operating Cash Flow	462	7.4	3.2	0	—

Note: “Q1 – Q3” indicates the share of fiscal years where all three quarterly facts (Q1, Q2, Q3) exist in XBRL. “Explicit Q4” indicates the share with a standalone fourth-quarter fact (period ≈ 90 days ending at fiscal year-end). “Comparable” is the count of firm-years for which both an explicit Q4

fact and the implied residual are available. “Match” reports the share of those comparable observations where the two values agree within 1% of the annual figure. Sample: 30 non-financial firms, all available fiscal years.

Missing risk-factor text in 10-Q. SEC Regulation S-K only requires firms to update risk factors in quarterly filings when “material changes” have occurred since the last 10-K. In practice, only about half the 10-Q filings in our pilot sample (15 firms, 90 filings) contained a legitimate risk-factor section. The rest had only a reference back to the annual report (Table 2.2). This held across sectors—technology, healthcare, consumer goods, energy—with few exceptions. Building Δ Risk, which enters both H1 and H2, is therefore not feasible at a quarterly frequency.

Chosen frequency. For these reasons—and following prior textual-analysis work based on 10-K filings (Blankespoor et al., 2020; Li, 2008; Loughran & McDonald, 2011)—we use an annual panel. The XBRL coverage comparison (Table A.1) and a consolidated summary (Table A.2) are in Appendix A.

Table 2.2: Textual Data Availability: 10-K vs. 10-Q Filings

	10-K	10-Q
Filings tested	44	90
<i>MD&A (Management’s Discussion and Analysis)</i>		
Section found (%)	93	100
Median word count	20,494	5,806
<i>Risk Factors</i>		
Substantive section found (%)	100	52
Boilerplate / reference only (%)	0	48
Median word count (substantive)	29,902	9,144

Note: “Found” requires >200 words extracted by regex-based section parser. “Substantive” for 10-Q excludes filings containing only a reference to the annual 10-K (e.g., “no material changes from those disclosed in our Annual Report”). MD&A corresponds to Item 7 in 10-K and Part I Item 2 in 10-Q; Risk Factors corresponds to Item 1A in 10-K and Part II Item 1A in 10-Q. Sample: 15 firms, most recent filings.

2.3 XBRL Tag Consistency Dilemma

A practical problem with the XBRL Company Facts API is that a single accounting concept can appear under different US-GAAP taxonomy tags. For example, operating cash flow may be reported as either `NetCashProvidedByUsedInOperatingActivities` (total, including discontinued operations) or `...ContinuingOperations` (continuing operations only). Net income may be tagged as `NetIncomeLoss`, `ProfitLoss`, or `IncomeLossFromContinuingOperations`. Stockholders’ equity can include or exclude the non-controlling interest. These tags overlap a lot but are not identical: mixing them all within a single firm’s time series introduces noise, while picking one tag globally would lose firms that consistently report under the other.

We handle this problem with a two-part *tag-harmonisation* process: (i) *equivalence groups* for tags that are economically interchangeable, and (ii) *per-firm tag locking* for tags that are not.

Equivalence groups. When two tags differ only in taxonomy version—not in their economic meaning—we let both enter a firm’s time series. Specifically, the two operating-cash-flow tags and the pair `NetIncomeLoss/ProfitLoss` are treated as interchangeable. The third net-income variant, `IncomeLossFromContinuingOperations`, is *not* included because it deliberately excludes discontinued operations and can differ materially.

To verify this classification, we queried the SEC EDGAR Company Facts API for all 550 sample firms and identified 6,113 firm-year periods in which both tags of a pair were reported simultaneously. The two overlap samples support treating the pairs as interchangeable:

- **Operating cash flow** (1,352 overlaps): 53.5% are numerically identical and the median difference scaled by total assets is practically zero. The 90th-percentile bias is only 1.5 pp.
- **Net income** (4,761 overlaps): 25.5% are numerically identical and the median ROA bias is 0.033 pp. 92.5% of observations lie below 1.0 pp.

The few cases with meaningful differences are already handled by our priority rule, which prefers the parent-company tag whenever both are reported.

Per-firm tag locking. For metrics where the alternative tags *are* economically distinct (stockholders' equity with vs. without NCI; `IncomeLossFromContinuingOperations` vs. `NetIncomeLoss`), we apply strict per-firm tag locking: we keep the tag each firm uses in the most fiscal years and set values from other tags to missing. Even if a firm switches its tag mid-sample, the locked series stays internally consistent at the cost of some discarded observations. This within-firm consistency is what our fixed-effects regressions require.

Combining equivalence groups and per-firm locking increases the complete-case rate for the five regression controls (*roa*, *leverage*, *asset_growth*, *current_ratio*, *log_total_assets*) from 78.9% to 92.7% of firm-year observations. This recovers 988 operating-cash-flow values and 292 net-income values that would otherwise be lost.

Residual limitation. The two-part process does not eliminate every source of cross-tag noise. About 4% of firms end up with a continuing-operations OCF tag paired with a total net-income tag. Ratios that combine the two—such as the auxiliary OCF-to-assets variable used in robustness—carry a small inconsistency. We accept this trade-off in exchange for better coverage. We winsorise these ratios at the 1st and 99th percentiles to clip the resulting extreme values, and note that the same kind of cross-tag mismatch is reported in comparable large-scale XBRL studies (Chychyla & Kogan, 2015).

Per-firm tag choices and the overlap diagnostic are reported in Appendix A (Table A.4).

2.4 How do we extract text from 10-K filings?

This study heavily depends on the textual sections of 10-K reports Item 1A *Risk Factors* and Item 7 *Management's Discussion and Analysis of Financial Condition and Results of Operations* (MD&A). This section describes how we find and extract those sections from the raw HTML filings on SEC EDGAR.

2.4.1 Source of textual data

All filings come directly from the SEC EDGAR via the Submissions API, which links each company’s Central Index Key (CIK) to the primary documents of its 10-K filings. From the panel of about 6,700 firm-year observations, we drop 10-K/A amendments. These typically consist of a cover page and a restated exhibit and do not contain the full narrative sections of the original 10-K making them unusable for further analysis. Financial data for amended fiscal years are kept, since the XBRL pipeline already picks up the amended figures when available (Section 2.3). Our extraction thus targets only original 10-K filings.

2.4.2 Section Identification

Unlike structured XBRL data, the textual parts of 10-K filings are hidden inside HTML with no standardised tagging. A raw filing can span across several thousands of lines, with section headers defined in many different ways: bold text, explicit `<font>` tags, inline uppercase text, or table-based layouts. Additionally, these headers are often mixed in with a table of contents (TOC), running headers, and inline cross-references (e.g. “see Item 1A”). These features make simple keyword search on plain text basically impossible, because it confuses TOC entries, page headers, and inline mentions with actual section boundaries.

To achieve the best possible results, we use an HTML extraction procedure with four stages:

1. **Header detection.** We walk through the HTML at the block level and test each element against a set of regex patterns for the target sections (e.g. a pattern that matches “*Item 1A*” together with its common variants). A candidate for correct section counts only if it meets at least one formatting rule: bold styling, ALL-CAPS text, or large font size (≥ 12 pt). This effectively filters out inline cross-references, which sit in normal body-text paragraphs with no special formatting, without discarding any valid headers. Elements that are themselves hyperlinks are also rejected, as these are almost always table-of-contents entries pointing to the real section.
2. **Position marking.** Each confirmed header is tagged in place before the document is flattened to plain text, so we keep track of where every header sits in the final text.
3. **Candidate selection.** For each target section, we keep only headers whose surrounding text contains an expected subtitle keyword (“risk” for Item 1A; “management” or “discussion” for Item 7) and discard any header having many other Item headers near it (within 300 characters in either direction), the typical signature of a table of contents. When several candidates remain, we pick the one followed by the longest stretch of text before the next section header.
4. **Section boundary.** The section text runs from the selected header to the first occurrence of the next expected Item header (Item 1B, 1C, or 2 for Item 1A; Item 7A or 8 for Item 7). Sections shorter than 200 words are treated as stubs and trigger a fallback procedure described in the next subsection.

2.4.3 Exhibit Fallback

A common practice in 10-K filings—especially older ones—is to replace an Item with a short stub such as “This information is incorporated by reference from Exhibit 13.” When the extracted section has fewer than 200 words and the text matches an incorporation-by-reference pattern, the parser finds the referenced exhibit number (defaulting to Exhibit 13, the annual report to shareholders) and downloads the matching HTML document from the filing index. The section identification procedure described in the previous subsection is then rerun on the exhibit. Because exhibits often skip the formal “Item X” prefix, the parser also matches patterns with titles only (e.g. “Risk Factors”, “Management’s Discussion and Analysis”) as a backup detection layer for both the main document and exhibits.

2.4.4 Post-Processing

After converting the HTML to plain text, several layout artefacts from the original filing remain. These could potentially harm later text analysis. Before computing any textual features, we perform 3 cleaning steps that remove three types of noise:

1. **Running page headers.** Many filings embed “*<page number> Table of Contents*” strings at page boundaries. A simple regular expression strips any line matching a one- to three-digit number followed by “Table of Contents.”
2. **Standalone page numbers.** After removing the full header pattern, some filings still contain page numbers on empty lines. These are also removed by deleting any line consisting solely of a one- to three-digit integer.
3. **Non-breaking spaces.** Non-breaking-space characters introduced by the original HTML markup are normalised to standard spaces, so that downstream word-count and dictionary-based measures are not affected by these artefacts.

The cleaning rules are intentionally conservative: they target only patterns that cannot realistically be part of meaningful text and are applied after section boundaries have already been set, so they do not interfere with header detection.

2.4.5 Coverage and Limitations

Because the quality of the extraction pipeline is crucial for downstream analysis, we validated it on two samples: the 50 most recent filings in the panel (fiscal year 2024) and, more importantly, a stratified random sample of 100 filings covering all fiscal years 2010–2024 (20 per era). On the recent sample, the parser successfully extracted both sections for all 50 filings (100%). On the cross-era sample, 97 of 100 filings yielded both sections. The three failures break down as follows:

- Two filings whose annual report exhibit was filed as a PDF rather than HTML, making

it inaccessible to an HTML parser;

- One filing whose 10-K used neither bold, large-font, nor uppercase cues for its section headers, leaving no formatting signal for the parser to use.

Neither failure mode is related to firm characteristics in any systematic way. On the full panel, we expect a coverage rate of roughly 97%, with missing values treated as missing in the textual variable construction.

LLM-assisted quality audit. Although the parser extracted 97/100 filings successfully, coverage alone says nothing about content quality. To check whether the extracted sections are actually the right ones, we ran an independent validation using a large language model (Google Gemini 2.5 Flash, accessed through its OpenAI-compatible API, temperature = 0 for deterministic results). Gilardi et al. (2023) show that LLM-based annotation matches or surpasses trained crowd-workers on text-classification tasks, which supports using a language model as an independent extraction validator. A stratified random sample of 100 filings was drawn—20 filings from each of five time periods (2010–2012, 2013–2015, 2016–2018, 2019–2021, 2022–2024). Both Item 1A and Item 7 were evaluated for every filing, giving 200 section-level evaluations in total. This design is in line with what Krippendorff (2019) recommends as reliable for content-analysis studies, and large enough that pass-rate estimates carry a margin of error of about ± 3 percentage points. The 100 filings also cover roughly 19 % of the 537-firm panel and span the full history of SEC filing formats.

For each filing, both extracted sections were sent to the model in a single prompt along with a structured evaluation rubric. The model evaluated each extracted section on five criteria—correct section identity, start-boundary accuracy, end-boundary accuracy, content quality (no residual HTML tags, page headers, or distorted data), and completeness (full narrative, not a stub or truncated fragment)—using a three-level grading scale: *Pass* (criterion fully met), *Minor* (small issue that would not significantly affect NLP analysis), and *Fail* (issue that would significantly affect NLP analysis). Additionally, we report two aggregate metrics: a strict rate (only Pass) and a lenient rate (Pass + Minor), each with Wilson 95 % confidence intervals (Agresti & Coull, 1998).

Table 2.3: LLM-assisted extraction validation results ($n = 200$ sections, 100 per section type).

Criterion	Strict	%	Lenient	%	95 % CI (lenient)
Correct section	192/200	96.0	192/200	96.0	[92.3 – 98.0 %]
Start boundary	191/200	95.5	193/200	96.5	[93.0 – 98.3 %]
End boundary	192/200	96.0	193/200	96.5	[93.0 – 98.3 %]
Content quality	191/200	95.5	195/200	97.5	[94.3 – 98.9 %]
Completeness	189/200	94.5	191/200	95.5	[91.7 – 97.6 %]
Overall (all five)	179/200	89.5	187/200	93.5	[89.2 – 96.2 %]

Table 2.3 shows the results. On every individual criterion, the lenient pass rate (Pass + Minor) is above 95%, confirming that the parser finds the right section with accurate boundaries and

produces clean, complete text. Even under the strict metric, the rates stay above 94% on each criterion. The overall lenient rate across all five criteria at once is 93.5% (187/200; 95 % CI: 89.2–96.2 %).

Of the 13 sections with at least one criteria resulting in Fail, five were complete extraction failures (empty output) caused by non-standard HTML layouts. Four had wrong section boundaries (content from an adjacent Item was included). Finally, the remaining four were flagged only for low word counts suggesting possible truncation, which we do not permit to enter the main analysis. Item 7 extractions do slightly better than Item 1A (lenient pass rates of 96% vs. 91%), most likely because Item 7 has an advantage in cleaner end boundary (Item 7A or Item 8), while Item 1A sometimes wrongly picks up text from the preceding Item 1.

Failures clearly cluster in older filings: 5 of the 40 sections from 2010–2012 were flagged. On the other hand, only 1 of 40 come from 2022–2024. This shows expected gradual standardisation of SEC filing HTML over the sample period.

Robustness of the LLM-based validation. Several choices in this validation procedure support the reliability of this approach. First, the validation is built around clearly defined, verifiable criteria—correct section identity, boundary accuracy, absence of HTML artefacts, and word-count thresholds—that leave little room for subjective interpretation and ambiguity. Second, each judgment covers a single document with a deterministic prompt. The model does not need to compare across filings or aggregate statistics. Third, manually checking the model’s Fail verdicts and reasons confirmed agreement in every case examined.

Relation to prior approaches. Our extraction strategy is inspired by Dyer et al. (2017), who document substantial evolution in 10-K disclosure—length, boilerplate, and structure—over time and the challenges this creates for large-sample studies. Compared with plain-text methods used in earlier work (Campbell et al., 2014; Li, 2008; Loughran & McDonald, 2011), our procedure avoids false positives from TOC entries and inline cross-references. Compared with XBRL-tagged textual blocks (e.g. `us-gaap:RiskFactorsTextBlock`), which became available with Inline XBRL adoption, our method covers a wider time span: structured textual tags are inconsistently filled before 2018, while the HTML-based approach works across the full 2010–2024 window.

3. Methodology

4. Results

4.1 Descriptive Statistics

Table 4.1 reports summary statistics for the regression sample. The dependent variable, post-filing annualised volatility, has a mean of about 0.33 and a standard deviation of 0.17, showing large cross-sectional differences in firm-level risk. The interquartile range (0.22 to 0.39) points to a right-skewed distribution, consistent with occasional spikes during market stress periods such as 2020.

Lagged volatility has a nearly identical distribution, as expected for a persistent series. Firm size, measured as $\ln(\text{Total Assets})$, ranges from mid-sized ($P25 \approx 22.1$) to large companies ($P75 \approx 24.2$). Leverage centres around 0.61 with moderate spread. ROA has a mean near 0.06, and asset growth averages about 0.12 with a long right tail driven by acquisitive firms.

Table 4.1: Summary Statistics

	Mean	SD	P25	Median	P75
Volatility _{t+1}	0.326	0.169	0.218	0.282	0.390
Lagged Volatility	0.325	0.173	0.217	0.279	0.387
$\ln(\text{Total Assets})$	23.134	1.483	22.131	23.146	24.212
Leverage	0.611	0.218	0.471	0.607	0.735
ROA	0.062	0.088	0.027	0.061	0.105
Asset Growth	0.117	0.251	0.003	0.060	0.144
Observations	6,299				
Firms	537				

Note: This table reports summary statistics for the regression sample. Volatility_{t+1} is the annualised standard deviation of daily log returns over the 365-day window following each 10-K filing date.

Leverage, ROA, and Asset Growth are winsorised at the 1st and 99th percentiles.

Table 4.2 presents pairwise Pearson correlations. Volatility and its lag are strongly positively correlated ($r = 0.61$), confirming persistence. Firm size correlates negatively with volatility ($r \approx -0.32$), in line with the well-known size–risk relationship. ROA also correlates negatively with volatility ($r \approx -0.33$), while leverage and asset growth show weaker univariate links. None of the pairwise correlations among the explanatory variables exceed 0.40 in absolute value, reducing multicollinearity concerns (variance inflation factors are below 1.4 for all regressors).

Table 4.2: Pearson Correlation Matrix

	(1)	(2)	(3)	(4)	(5)	(6)
(1) Volatility _{t+1}	1					
(2) Lagged Volatility	0.61***	1				
(3) ln(Total Assets)	-0.32***	-0.33***	1			
(4) Leverage	-0.07***	-0.05***	0.21***	1		
(5) ROA	-0.33***	-0.38***	0.07***	-0.06***	1	
(6) Asset Growth	0.17***	0.17***	-0.13***	-0.11***	-0.05***	1

Note: This table reports pairwise Pearson correlation coefficients. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

4.2 Baseline: Financial Determinants of Volatility

This section establishes the financial-only benchmark against which the text and divergence models presented later in this chapter are evaluated. Under the null hypothesis H0, narrative disclosures add no predictive power once standard accounting fundamentals and volatility persistence are accounted for. The adjusted R^2 of the specification reported in Column (3) of Table 4.3 therefore defines the bar that any text-based extension must clear.

Table 4.3 reports OLS estimates of post-filing annualised volatility on progressively richer sets of financial controls. All three specifications absorb two-digit SIC industry (49 groups) and fiscal-year fixed effects, so the coefficients are identified from within-industry, within-year variation. Standard errors are clustered at the firm level (Petersen, 2009) to accommodate within-firm serial correlation, which is amplified by the partial overlap of the 365-day return windows used to construct $\sigma_{i,t+1}$.

Volatility persistence. Column (1) includes only lagged volatility ($\sigma_{i,t}$) alongside the fixed effects. The coefficient of 0.67 ($t = 33.5$) confirms the well-documented persistence of firm-level return volatility: a one percentage-point rise in last year’s annualised volatility maps into about a two-thirds percentage-point rise the following year. This single regressor, combined with industry and year fixed effects, already explains about 63% of the cross-sectional and time-series variation in post-filing volatility, showing that most of the predictable component of risk is mechanical.

Firm size and capital structure. Column (2) adds firm size, measured as the natural logarithm of total assets, and book leverage (total liabilities over total assets). Size enters with a negative and highly significant coefficient (-0.017 , $t = -7.4$): larger firms exhibit lower future volatility, consistent with the size–risk relationship documented for U.S. equities by Chi and Su (2017) and Campbell et al. (2014). Leverage is economically negligible and statistically indistinguishable from zero ($t = 0.17$); the industry fixed effects appear to absorb most of the

Table 4.3: Baseline Volatility Determinants

	(1)	(2)	(3)
Lagged Volatility	0.6733*** (33.53)	0.6116*** (30.19)	0.5501*** (20.73)
ln(Total Assets)		−0.0167*** (−7.36)	−0.0166*** (−7.50)
Leverage		0.0017 (0.17)	−0.0002 (−0.02)
ROA			−0.2240*** (−7.95)
Asset Growth			0.0260*** (3.14)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	6,299	6,299	6,299
Firms	537	537	537
Adj. R^2	0.630	0.643	0.654

Note: This table reports OLS estimates of post-filing annualised volatility on financial determinants.

The dependent variable is the annualised standard deviation of daily log returns over the 365-day window following each 10-K filing date. All specifications include two-digit SIC industry (49 groups) and fiscal-year fixed effects. t -statistics, reported in parentheses, are based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

cross-sectional dispersion in capital structure, leaving little residual variation through which leverage can operate.

Profitability and growth. The full specification in Column (3) adds return on assets and asset growth. ROA carries a large negative coefficient (-0.224 , $t = -7.9$), confirming that profitable firms face lower subsequent volatility, in line with the earnings-quality intuition of Francis et al. (2005). Asset growth enters positively ($+0.026$, $t = 3.1$): rapid expansion is associated with elevated future uncertainty, consistent with the operational-risk channel. Adding these two regressors raises the adjusted R^2 from 0.643 to 0.654—a modest but statistically meaningful improvement that refines, rather than dominates, the persistence-and-size benchmark.

A noteworthy by-product of the full specification is that the lagged volatility coefficient declines from 0.67 in Column (1) to 0.55 ($t = 20.7$) in Column (3). Roughly one-fifth of what looked like raw persistence is in fact mediated by slower-moving fundamentals—size, profitability, and growth—that are themselves persistent. Lagged volatility nonetheless remains by a wide margin the most informative single regressor in the model.

Benchmark for textual analysis. The full baseline model explains 65.4% of the variation in post-filing volatility. This adjusted R^2 is the benchmark against which the text model and the divergence model are evaluated, both through incremental R^2 and through joint F -tests on the textual coefficients. The bar is high: financial fundamentals together with volatility persistence already capture most of the predictable variation in firm-level risk, leaving narrative disclosures to compete for what remains. Establishing this benchmark is itself a contribution, because it isolates the share of post-filing volatility that any text-based mechanism must explain beyond what is mechanically traceable to the balance sheet.

5. Robustness

Conclusion

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Appendices

A. Variable Definitions and Data Pipeline

This appendix collects supplementary tables referenced in the main text. Section A.1 presents the diagnostics behind the choice of annual filing frequency (Section 2.2). Section A.2 reports the per-firm XBRL tag assignments from the tag-harmonisation procedure (Section 2.3).

A.1 Filing Frequency Diagnostics

The following tables supplement the discussion in Section 2.2.

Table A.1: XBRL Financial Metric Coverage: Annual (10-K) vs. Quarterly (10-Q)

Metric	Annual (%)	Quarterly (%)
assets_current	87.0	91.1
eps_diluted	89.4	92.7
liabilities_current	87.0	91.1
net_income	93.6	97.3
operating_cash_flow	89.2	35.8
operating_income	76.9	80.5
stockholders_equity	98.7	94.9
total_assets	89.9	93.3
total_liabilities	63.7	66.5
Median periods per firm	20	52
Firms with data	30	30

Note: Coverage shows the percentage of firm-periods with non-missing values for each metric. Annual data is extracted from 10-K filings; quarterly data from 10-Q filings (Q1–Q3 only). Sample: 30 non-financial operating companies from the top 200 by market capitalisation.

Table A.2: Annual vs. Quarterly Approach: Summary Comparison

Criterion	Annual (10-K)	Quarterly (10-Q)
Filing type	10-K	10-Q (Q1–Q3 only)
MD&A availability	93% (Item 7)	100% (Part I, Item 2)
Risk Factors availability	100% (Item 1A)	~52% substantive
Risk Factors content	Median ~30,000 words	Median ~9,000 words (substantive); 48% boilerplate
Q4 financial data	Included in 10-K	Unreliable / absent
Explicit Q4 XBRL facts	N/A	3–43% of firm-years
Implied Q4 mismatch rate	N/A	~11% of comparisons
Cash-flow quarterly coverage	N/A	7% of firm-years
NLP variable: Δ Risk	Feasible	Limited (52% substantive)
NLP variable: Δ Sentiment	Feasible	Feasible (noisier)
NLP variable: TextSim	YoY (standard)	QoQ (unclear construct)
Literature alignment	Standard	Uncommon
Expected observations	~6,400 firm-years	~19,000 firm-quarters

Note: Summary based on diagnostic tests of 30 non-financial companies (XBRL, Q4 tests) and 15 companies (textual tests). Risk Factors in 10-Q filings are required by SEC Regulation S-K Item 1A only when there are “material changes” from the most recent 10-K; roughly half of companies include substantive risk disclosures, while the rest provide only a boilerplate reference. See Tables A.1, 2.1, and 2.2 for detailed results.

Table A.3: Per-Company 10-Q Textual Content (Most Recent 6 Filings)

Ticker	Filings	MD&A Found	MD&A Med. Words	Risk Subst.	Risk Boilerplate
AAPL	6	6	2,781	3	3
AMZN	6	6	6,291	6	0
AVGO	6	6	4,778	6	0
COST	6	6	4,979	0	6
GOOGL	6	6	8,096	3	3
JNJ	6	6	7,587	0	6
LLY	6	6	4,994	0	6
MA	6	6	7,300	6	0
META	6	6	7,756	6	0
MSFT	6	6	7,981	0	6
NVDA	6	6	5,156	6	0
TSLA	6	6	5,582	2	4
V	6	6	5,043	0	6
WMT	6	6	7,142	3	3
XOM	6	6	6,197	6	0

Note: “MD&A Found” and “Risk Subst.” count filings with >200 substantive words. “Risk Boilerplate” counts filings where the risk-factors section contains only a reference to the annual 10-K.

A.2 XBRL Tag Provenance

Table A.4 reports, for each metric with multiple candidate XBRL tags, the number and share of firms locked to each tag by the per-firm tag-locking procedure described in Section 2.3.

Table A.4: Per-firm XBRL tag distribution

Metric	Locked XBRL tag	Firms	%
Net income	NetIncomeLoss	502	91.6
	ProfitLoss	45	8.2
	IncomeLossFromContinuingOperations	1	0.2
Operating cash flow	NetCashProvidedByUsedInOperatingActivities	530	96.7
	...OperatingActivitiesContinuingOperations	18	3.3
Stockholders' equity	StockholdersEquityIncl...NoncontrollingInterest	318	58.6
	StockholdersEquity	225	41.4

Notes. For each (firm, metric) pair, the tag with the highest within-firm year-count is selected; values from other tags are set to missing. Firm counts are based on up to 548 firms in the tag-locking stage; the stockholders' equity row sums to 543 because five firms report no equity tag at all. See Section 2.3 for the rationale.

B. Documentation of AI Tool Usage

This appendix documents the use of artificial intelligence tools in preparing this thesis, as required by the university.

Extraction Validation (Section 2.4.5)

Tool. Google Gemini 2.5 Flash (accessed via the OpenAI-compatible Gemini API endpoint), with temperature set to 0 for deterministic output.

Purpose. An independent, automated quality audit of the SEC 10-K section extraction pipeline. The model evaluated 200 extracted sections (100 Item 1A, 100 Item 7) against a structured five-criterion rubric (correct section, start boundary, end boundary, content quality, completeness) on a three-level scale (Pass / Minor / Fail).

Contribution and verification. The evaluation rubric, sampling strategy, prompt design, and statistical analysis (Wilson confidence intervals) were all designed by the author. The model's only role was applying the rubric to each extracted section. All Fail verdicts were manually spot-checked against the original SEC filings and confirmed to be correct. Results are in Table 2.3.

Code Development

Tool. AI-powered code completion (integrated in VS Code).

Purpose. Code completion during the development of the data-pipeline scripts that build the panel and extract the 10-K text sections.

Contribution and verification. All generated code was reviewed, tested, and modified by the author. The overall architecture, variable definitions, and analytical choices are the author's own.

Writing Assistance

Tool. AI-powered writing assistant (integrated in VS Code).

Purpose. Grammar and style suggestions for English-language prose in the thesis text.

Contribution and verification. All substantive content, argumentation, and interpretation are the author's own. The tool was used only for language polishing; every suggestion was reviewed before acceptance.